V. PAVAN SAI

AI/ML Intern – Raritone Summer Internship 2026

Project Report: AI-Powered Virtual Try-On System for Fashion E-Commerce

Introduction

This project focused on researching AI-powered Virtual Try-On systems, Computer Vision, Human Pose Estimation, Cloth Segmentation, Recommendation Systems, and AR/VR technologies for fashion e-commerce applications.

Objectives

Study body detection, pose estimation, cloth warping, virtual try-on systems, recommendation systems, datasets, AI models, and future improvements for Raritone.

Research Areas Covered

Human Body Detection, Human Pose Estimation, Body Measurement Analysis, Cloth Segmentation & Human Parsing, 2D Virtual Try-On, 3D Virtual Try-On, AI Recommendation Systems, AR/VR Technologies.

Models Explored

MediaPipe, OpenPose, U2Net, Detectron2, HR-VITON, DensePose, Hybrid Recommendation Models.

Datasets Collected

DeepFashion, DeepFashion2, VITON, VITON-HD, MPV Dataset, Human Pose Datasets, Body Segmentation Datasets.

2D Virtual Try-On Findings

2D systems provide better accessibility, faster processing, lower computational requirements, and improved online shopping experiences.

3D Virtual Try-On Findings

3D systems provide realistic fitting, avatar generation, cloth simulation, motion tracking, and immersive shopping experiences.

AI Fashion Recommendation System

Studied Content-Based Filtering, Collaborative Filtering, and Hybrid Recommendation Systems to improve personalization and product discovery.

Complete AI Workflow

User Upload → Body Detection → Pose Estimation → Body Segmentation → Measurement Analysis → Garment Mapping → Virtual Try-On → Fashion Recommendation → Final Output

Challenges Faced

Clothing Alignment Issues, Background Noise, Different Body Shapes, Real-Time Performance, Dataset Limitations, Recommendation Accuracy, High Computing Cost.

Solutions Implemented

TPS and GMM Warping, U2Net Segmentation, DensePose Fitting, MediaPipe Optimization, Hybrid Recommendation Model, Cloud Infrastructure.

Suggestions for Raritone

Implement AI-based size recommendations, personalized fashion recommendations, customer behavior analytics, AR-based live try-on, and 3D avatar shopping experiences.

Future Scope

Real-time Virtual Try-On, Advanced 3D Virtual Try-On, AI Personal Stylist, AR-Based Fashion Shopping, Improved Body Measurement Estimation, Enhanced Recommendation Systems, Mobile Integration, Scalable AI Infrastructure.

Key Learnings

Computer Vision, Human Pose Estimation, Human Parsing, Cloth Warping, Recommendation Systems, AR/VR Technologies, Deep Learning Models, Dataset Evaluation, Fashion AI Applications.

Conclusion

The project successfully explored AI-powered fashion technologies and demonstrated how Virtual Try-On systems and AI recommendations can improve customer engagement, personalization, and online shopping experiences.